

Ali Ahmed Mohamed Hassan

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Computer Science student at the German International University (GIU) with hands-on experience in Python, Java, SQL, machine learning, and cryptography. Demonstrated ability to build end-to-end solutions across AI-powered web platforms, supervised and unsupervised ML models, data analysis pipelines, security systems, and full-stack database applications. Eager to contribute in a software engineering or data science internship role.

CORE COMPETENCIES

Python · Machine Learning · Data Analysis · SQL & Database Design · Java OOP · Full-Stack Web Development

EDUCATION

B.Sc. Computer Science (In Progress) – German International University (GIU) Cairo, Egypt 2024 – Present
Faculty of Informatics and Computer Science | Grade: Very Good

- Relevant coursework: Data Structures, Algorithms, Databases, Software Engineering, OOP

American High School Diploma – Nefertari American International School Cairo, Egypt 2024

- GPA: 4.0 | EST I: 1320 | EST II: 1160 | Graduated with distinction

PROJECTS

AI-Powered Internship Finder – Full-Stack Web Application 2026

- Built a JavaScript web platform using AI to match students with internships based on skills and preferences
- Integrated an AI recommendation engine delivering personalized listings with real-time dynamic filtering

Metabolic Syndrome Risk Classifier – Machine Learning – Supervised 2026

- Designed preprocessing pipelines (imputation, normalization, encoding) on a messy clinical dataset with missing values
- Benchmarked Logistic Regression, Random Forest, and SVM; evaluated via cross-validation, confusion matrices, and F1-score

Unsupervised Clustering Model – Machine Learning – Unsupervised 2026

- Implemented K-Means and hierarchical clustering to segment data; applied PCA for dimensionality reduction and visualized cluster quality

Football Player Performance Analysis – Data Analysis 2026

- Analyzed player metrics (goals, assists, pass accuracy, defensive contributions) across positions and seasons using Pandas and Matplotlib
- Applied statistical techniques to identify top performers and visualize performance trends over time

HR Database System – SQL · Web Development 2025

- Designed a normalized relational database (employees, departments, salaries, contracts) with ER modelling and FK constraints
- Built a web interface for full CRUD operations, report generation, and role-based data access

Jackroo – Strategy Board Game – Java · OOP 2025

- Developed full Jackroo game in Java with all rules, turn management, and win-condition logic using OOP design patterns

3D Game – Unity · C# 2026

- Built a 3D interactive Unity game with physics-based mechanics, scene management, and optimized rendering performance

EXTRACURRICULAR ACTIVITIES

Volunteer – El Ganan Orphanage Cairo, Egypt

2025 – Present

- Contributed regular community service supporting children's educational and recreational activities

TECHNICAL SKILLS & LANGUAGES

Technical Programs: Python (Advanced), Java (Advanced), C++ (Intermediate), JavaScript (Intermediate), SQL (Advanced), C# (Intermediate)

Tools & Frameworks: Hugging Face , NodeJs, Unity, MySQL, Git & GitHub, Visual Studio Code

Domains: Machine Learning, Data Analysis, Cryptography, Database Design, Full-Stack Web Development

Languages: Arabic (Native), English (Fluent), French (Conversational), German (A2)

Portfolio: <https://aliahmedd4.github.io/>